RAVELOMANANA Anja Rino Stive

L2 INFO

182I25

BD_AIR_MAROC

Création base de données

```
MariaDB [(none)]> CREATE DATABASE BD_AIR_MAROC;
Query OK, 1 row affected (0.001 sec)
```

Création des tables

```
MariaDB [BD_AIR_MAROC]> CREATE TABLE Pilotes (
    ->     NUMPIL INT AUTO_INCREMENT PRIMARY KEY,
    ->     NOMPIL VARCHAR(70) NOT NULL,
    ->     Ville VARCHAR(50) NOT NULL,
    ->     Salaire DECIMAL(10, 2) NOT NULL
    -> ) AUTO_INCREMENT = 1;
Query OK, 0 rows affected (0.011 sec)

MariaDB [BD_AIR_MAROC]> CREATE TABLE Avion (
    ->     NUMAV INT AUTO_INCREMENT PRIMARY KEY,
    ->     Nomav VARCHAR(50) NOT NULL,
    ->     Capacite INT NOT NULL,
    ->     Ville VARCHAR(50) NOT NULL
    -> ) AUTO_INCREMENT = 100;
Query OK, 0 rows affected (0.017 sec)

MariaDB [BD_AIR_MAROC]> CREATE TABLE Vol (
    ->     Numvol VARCHAR(10) PRIMARY KEY,
    ->     NUMPIL INT NOT NULL,
    ->     NUMAV INT NOT NULL,
    ->     Ville_dep VARCHAR(50) NOT NULL,
    ->     Ville_arr VARCHAR(50) NOT NULL,
    ->     H_dep TIME NOT NULL,
    ->     H_arr TIME NOT NULL,
    ->     FOREIGN KEY (NUMPIL) REFERENCES Pilotes(NUMPIL),
    ->     FOREIGN KEY (NUMAV) REFERENCES Avion(NUMAV)
    -> );
Query OK, 0 rows affected (0.045 sec)
```

Insertion des données

Données pilotes :

```
MariaDB [BD_AIR_MAROC]> INSERT INTO Pilotes (nompil, ville, salaire) VALUES
    -> ('Alami', 'Marrakech', 12500), ('Benani', 'Meknes', 30000), ('Chraibi', 'Guelmim',
    -> 28000), ('Drissi', 'Casablanca', 13500), ('El Fassi', 'Rabat', 29500),
    -> ('Ghazali', 'Fes', 12000), ('Haddi', 'Tanger', 25500), ('Idrissi', 'Agadir', 12000),
    -> ('Jebli', 'Oujda', 12000), ('Kabbaj', 'Kenitra', 30000),
    -> ('Lahlou', 'Tetouan', 12000), ('Mansouri', 'Safi', 13000), ('Naceri', 'El Jadida',
    -> 19500), ('Ouazzani', 'Nador', 13200), ('Qasmi', 'Settat', 26400),
    -> ('Radi', 'Marrakech', 28000), ('Saidi', 'Meknes', 13900), ('Tazi', 'Guelmim', 18200),
    -> ('Uali', 'Casablanca', 10800), ('Vafi', 'Rabat', 24500),
    -> ('Wadi', 'Fes', 19200), ('Xali', 'Tanger', 30000), ('Yacoubi', 'Agadir', 18200),
    -> ('Zouhair', 'Oujda', 30000), ('Amrani', 'Kenitra', 27500),
    -> ('Bekkali', 'Tetouan', 21200), ('Cheddadi', 'Safi', 30000), ('Dahbi', 'El Jadida',
    -> 29700), ('Filali', 'Nador', 23400), ('Gouni', 'Settat', 16700),
    -> ('Habibi', 'Marrakech', 30000), ('Iraqi', 'Meknes', 12000), ('Jouahri', 'Guelmim',
    -> 18400), ('Kadir', 'Casablanca', 20500), ('Lamrani', 'Rabat', 24800),
    -> ('Mernissi', 'Fes', 12000), ('Naji', 'Tanger', 30000), ('Omari', 'Agadir', 12000),
    -> ('Perrin', 'Oujda', 26000), ('Rami', 'Kenitra', 17800),
    -> ('Slaoui', 'Tetouan', 29400), ('Toumi', 'Safi', 30000), ('Utmani', 'El Jadida',
    -> 19900), ('Vadi', 'Nador', 23600), ('Wafi', 'Settat', 27000),
    -> ('Younes', 'Marrakech', 30000), ('Ziani', 'Meknes', 12000), ('Abidi', 'Guelmim',
    -> 18600), ('Bari', 'Casablanca', 30000), ('Chafi', 'Rabat', 15000);
Query OK, 50 rows affected (0.009 sec)
Records: 50  Duplicates: 0  Warnings: 0
```

Données Avion :

```
MariaDB [BD_AIR_MAROC]> INSERT INTO Avion (nomav, capacite, ville) VALUES
    -> ('Boeing 737', 180, 'Casablanca'), ('Airbus A320', 150, 'Marrakech'), ('Boeing 747',
    -> 400, 'Rabat'), ('Airbus A350', 320, 'Fes'), ('ATR 72', 160, 'Agadir'),
    -> ('Boeing 777', 350, 'Tanger'), ('Airbus A330', 280, 'Meknes'), ('Embraer 190', 170,
    -> 'Oujda'), ('Boeing 787', 310, 'Guelmim'), ('Airbus A321', 200, 'Kenitra'),
    -> ('Boeing 737', 185, 'Tetouan'), ('Airbus A320', 155, 'Safi'), ('Boeing 747', 390,
    -> 'El Jadida'), ('Airbus A350', 310, 'Nador'), ('ATR 72', 165, 'Settat'),
    -> ('Boeing 777', 340, 'Casablanca'), ('Airbus A330', 270, 'Marrakech'), ('Embraer
    '> 190', 175, 'Rabat'), ('Boeing 787', 300, 'Fes'), ('Airbus A321', 210, 'Agadir'),
    -> ('Boeing 737', 190, 'Tanger'), ('Airbus A320', 160, 'Meknes'), ('Boeing 747', 380,
    -> 'Oujda'), ('Airbus A350', 300, 'Guelmim'), ('ATR 72', 170, 'Kenitra'),
    -> ('Boeing 777', 330, 'Tetouan'), ('Airbus A330', 260, 'Safi'), ('Embraer 190', 180,
    -> 'El Jadida'), ('Boeing 787', 290, 'Nador'), ('Airbus A321', 220, 'Settat'),
    -> ('Boeing 737', 195, 'Casablanca'), ('Airbus A320', 165, 'Marrakech'), ('Boeing 747',
    -> 370, 'Rabat'), ('Airbus A350', 290, 'Fes'), ('ATR 72', 175, 'Agadir'),
    -> ('Boeing 777', 320, 'Tanger'), ('Airbus A330', 250, 'Meknes'), ('Embraer 190', 185,
    -> 'Oujda'), ('Boeing 787', 280, 'Guelmim'), ('Airbus A321', 230, 'Kenitra'),
    -> ('Boeing 737', 200, 'Tetouan'), ('Airbus A320', 170, 'Safi'), ('Boeing 747', 360,
    -> 'El Jadida'), ('Airbus A350', 280, 'Nador'), ('ATR 72', 180, 'Settat'),
    -> ('Boeing 777', 310, 'Casablanca'), ('Airbus A330', 240, 'Marrakech'), ('Embraer
    '> 190', 190, 'Rabat'), ('Boeing 787', 270, 'Fes'), ('Airbus A321', 240, 'Agadir');
Query OK, 50 rows affected (0.008 sec)
Records: 50  Duplicates: 0  Warnings: 0
```

Données VOL :

```
MariaDB [BD_AIR_MAROC]> INSERT INTO Vol (numvol, numpil, NUMAV, ville_dep, ville_arr, h_dep, h_arr) VALUES
    -> ('V103', 12, 102, 'Agadir', 'Tanger', '14:00', '15:45'),
    -> ('V104', 5, 110, 'Fes', 'Oujda', '07:15', '08:00'),
    -> ('V105', 50, 140, 'Guelmim', 'Meknes', '16:00', '18:30'),
    -> ('V106', 11, 120, 'Casablanca', 'Rabat', '19:00', '19:45'),
    -> ('V107', 25, 104, 'Tetouan', 'Casablanca', '11:00', '12:15'),
    -> ('V108', 2, 103, 'Meknes', 'Fes', '08:30', '09:00'),
    -> ('V109', 2, 102, 'Fes', 'Meknes', '10:00', '10:30'),
    -> ('V110', 3, 102, 'Guelmim', 'Agadir', '13:00', '14:00'),
    -> ('V111', 4, 103, 'Casablanca', 'Tanger', '15:00', '16:30'),
    -> ('V112', 6, 104, 'Fes', 'Marrakech', '09:00', '10:30'),
    -> ('V113', 7, 106, 'Tanger', 'Oujda', '12:00', '13:15'),
    -> ('V114', 8, 107, 'Agadir', 'Safi', '17:00', '18:00'),
    -> ('V115', 9, 108, 'Oujda', 'Nador', '08:00', '08:45'),
    -> ('V116', 10, 109, 'Kenitra', 'Settat', '14:00', '15:00'),
    -> ('V117', 11, 111, 'Tetouan', 'Rabat', '10:00', '11:15'),
    -> ('V118', 13, 112, 'El Jadida', 'Casablanca', '16:00', '17:00'),
    -> ('V119', 14, 113, 'Nador', 'Fes', '07:00', '08:00'),
    -> ('V120', 15, 114, 'Settat', 'Meknes', '18:00', '19:30'),
    -> ('V121', 16, 115, 'Marrakech', 'Guelmim', '09:30', '10:45'),
    -> ('V122', 17, 116, 'Meknes', 'Tanger', '11:00', '12:30'),
    -> ('V123', 18, 117, 'Guelmim', 'Casablanca', '15:00', '16:45'),
    -> ('V124', 19, 118, 'Casablanca', 'Agadir', '08:00', '09:30'),
    -> ('V125', 20, 119, 'Rabat', 'Oujda', '13:00', '14:30'),
    -> ('V126', 21, 121, 'Fes', 'Kenitra', '10:00', '11:00'),
    -> ('V127', 22, 122, 'Tanger', 'Tetouan', '14:00', '14:45'),
    -> ('V128', 23, 123, 'Agadir', 'Safi', '17:00', '18:15'),
    -> ('V129', 24, 124, 'Oujda', 'El Jadida', '09:00', '10:30'),
    -> ('V130', 26, 125, 'Tetouan', 'Nador', '12:00', '13:00'),
    -> ('V131', 27, 126, 'Safi', 'Settat', '15:00', '16:15'),
    -> ('V132', 28, 127, 'El Jadida', 'Marrakech', '08:00', '09:45'),
    -> ('V133', 29, 128, 'Nador', 'Meknes', '11:00', '12:45'),
    -> ('V134', 30, 129, 'Settat', 'Guelmim', '16:00', '17:45'),
    -> ('V135', 31, 130, 'Marrakech', 'Casablanca', '07:00', '08:00'),
    -> ('V136', 32, 131, 'Meknes', 'Rabat', '10:00', '11:15'),
    -> ('V137', 33, 132, 'Guelmim', 'Fes', '14:00', '15:45'),
    -> ('V138', 34, 133, 'Casablanca', 'Tanger', '18:00', '19:30'),
    -> ('V139', 35, 134, 'Rabat', 'Agadir', '09:00', '10:45'),
    -> ('V140', 36, 135, 'Fes', 'Oujda', '12:00', '13:00'),
    -> ('V141', 37, 136, 'Tanger', 'Kenitra', '15:00', '16:00'),
    -> ('V142', 38, 137, 'Agadir', 'Tetouan', '08:00', '10:00'),
    -> ('V143', 39, 138, 'Oujda', 'Safi', '11:00', '13:00'),
    -> ('V144', 40, 139, 'Kenitra', 'El Jadida', '16:00', '17:30'),
    -> ('V145', 41, 140, 'Tetouan', 'Nador', '07:30', '08:30'),
    -> ('V146', 42, 141, 'Safi', 'Settat', '14:00', '15:15'),
    -> ('V147', 43, 142, 'El Jadida', 'Marrakech', '19:00', '20:45'),
    -> ('V148', 44, 143, 'Nador', 'Meknes', '10:00', '11:45'),
    -> ('V149', 45, 144, 'Settat', 'Guelmim', '13:00', '14:45'),
    -> ('V150', 10, 115, 'Marrakech', 'Casablanca', '21:00', '22:00');
Query OK, 48 rows affected (0.011 sec)
Records: 48  Duplicates: 0  Warnings: 0
```

1. Avions avec capacité > 350 :

```
MariaDB [BD_AIR_MAROC]> SELECT * FROM AVION WHERE CAPACITE > 350;
+-------+------------+----------+-----------+
| NUMAV | Nomav      | Capacite | Ville     |
+-------+------------+----------+-----------+
|   102 | Boeing 747 |      400 | Rabat     |
|   112 | Boeing 747 |      390 | El Jadida |
|   122 | Boeing 747 |      380 | Oujda     |
|   132 | Boeing 747 |      370 | Rabat     |
|   142 | Boeing 747 |      360 | El Jadida |
+-------+------------+----------+-----------+
5 rows in set (0.000 sec)
```

2. Avions à Marrakech :

```
MariaDB [BD_AIR_MAROC]> SELECT NUMAV, NOMAV FROM AVION WHERE VILLE = 'Marrakech';
+-------+-------------+
| NUMAV | NOMAV       |
+-------+-------------+
|   101 | Airbus A320 |
|   116 | Airbus A330 |
|   131 | Airbus A320 |
|   146 | Airbus A330 |
+-------+-------------+
4 rows in set (0.000 sec)
```

3. Pilotes en service et villes de départ :

```
MariaDB [BD_AIR_MAROC]> SELECT DISTINCT NUMPIL, VILLE_DEP FROM VOL;
+--------+------------+
| NUMPIL | VILLE_DEP  |
+--------+------------+
|     12 | Casablanca |
|     10 | Marrakech  |
|     12 | Agadir     |
|      5 | Fes        |
|     50 | Guelmim    |
|     11 | Casablanca |
|     25 | Tetouan    |
|      2 | Meknes     |
|      2 | Fes        |
|      3 | Guelmim    |
|      4 | Casablanca |
|      6 | Fes        |
|      7 | Tanger     |
|      8 | Agadir     |
|      9 | Oujda      |
|     10 | Kenitra    |
|     11 | Tetouan    |
|     13 | El Jadida  |
|     14 | Nador      |
|     15 | Settat     |
|     16 | Marrakech  |
|     17 | Meknes     |
|     18 | Guelmim    |
|     19 | Casablanca |
|     20 | Rabat      |
|     21 | Fes        |
|     22 | Tanger     |
|     23 | Agadir     |
|     24 | Oujda      |
|     26 | Tetouan    |
|     27 | Safi       |
|     28 | El Jadida  |
|     29 | Nador      |
|     30 | Settat     |
|     31 | Marrakech  |
|     32 | Meknes     |
|     33 | Guelmim    |
|     34 | Casablanca |
|     35 | Rabat      |
|     36 | Fes        |
|     37 | Tanger     |
|     38 | Agadir     |
|     39 | Oujda      |
|     40 | Kenitra    |
|     41 | Tetouan    |
|     42 | Safi       |
|     43 | El Jadida  |
|     44 | Nador      |
|     45 | Settat     |
+--------+------------+
49 rows in set (0.000 sec)
```

4. Toutes les informations sur les pilotes :

```
MariaDB [BD_AIR_MAROC]> SELECT * FROM PILOTES;
+--------+----------+------------+----------+
| NUMPIL | NOMPIL   | Ville      | Salaire  |
+--------+----------+------------+----------+
|      1 | Alami    | Marrakech  | 12500.00 |
|      2 | Benani   | Meknes     | 30000.00 |
|      3 | Chraibi  | Guelmim    | 28000.00 |
|      4 | Drissi   | Casablanca | 13500.00 |
|      5 | El Fassi | Rabat      | 29500.00 |
|      6 | Ghazali  | Fes        | 12000.00 |
|      7 | Haddi    | Tanger     | 25500.00 |
|      8 | Idrissi  | Agadir     | 12000.00 |
|      9 | Jebli    | Oujda      | 12000.00 |
|     10 | Kabbaj   | Kenitra    | 30000.00 |
|     11 | Lahlou   | Tetouan    | 12000.00 |
|     12 | Mansouri | Safi       | 13000.00 |
|     13 | Naceri   | El Jadida  | 19500.00 |
|     14 | Ouazzani | Nador      | 13200.00 |
|     15 | Qasmi    | Settat     | 26400.00 |
|     16 | Radi     | Marrakech  | 28000.00 |
|     17 | Saidi    | Meknes     | 13900.00 |
|     18 | Tazi     | Guelmim    | 18200.00 |
|     19 | Uali     | Casablanca | 10800.00 |
|     20 | Vafi     | Rabat      | 24500.00 |
|     21 | Wadi     | Fes        | 19200.00 |
|     22 | Xali     | Tanger     | 30000.00 |
|     23 | Yacoubi  | Agadir     | 18200.00 |
|     24 | Zouhair  | Oujda      | 30000.00 |
|     25 | Amrani   | Kenitra    | 27500.00 |
|     26 | Bekkali  | Tetouan    | 21200.00 |
|     27 | Cheddadi | Safi       | 30000.00 |
|     28 | Dahbi    | El Jadida  | 29700.00 |
|     29 | Filali   | Nador      | 23400.00 |
|     30 | Gouni    | Settat     | 16700.00 |
|     31 | Habibi   | Marrakech  | 30000.00 |
|     32 | Iraqi    | Meknes     | 12000.00 |
|     33 | Jouahri  | Guelmim    | 18400.00 |
|     34 | Kadir    | Casablanca | 20500.00 |
|     35 | Lamrani  | Rabat      | 24800.00 |
|     36 | Mernissi | Fes        | 12000.00 |
|     37 | Naji     | Tanger     | 30000.00 |
|     38 | Omari    | Agadir     | 12000.00 |
|     39 | Perrin   | Oujda      | 26000.00 |
|     40 | Rami     | Kenitra    | 17800.00 |
|     41 | Slaoui   | Tetouan    | 29400.00 |
|     42 | Toumi    | Safi       | 30000.00 |
|     43 | Utmani   | El Jadida  | 19900.00 |
|     44 | Vadi     | Nador      | 23600.00 |
|     45 | Wafi     | Settat     | 27000.00 |
|     46 | Younes   | Marrakech  | 30000.00 |
|     47 | Ziani    | Meknes     | 12000.00 |
|     48 | Abidi    | Guelmim    | 18600.00 |
|     49 | Bari     | Casablanca | 30000.00 |
|     50 | Chafi    | Rabat      | 15000.00 |
+--------+----------+------------+----------+
50 rows in set (0.000 sec)
```

5. Pilotes à Meknès avec salaire > 20000 :

```
MariaDB [BD_AIR_MAROC]> SELECT NOMPIL FROM PILOTES WHERE VILLE = 'Meknès' AND SALAIRE > 20000;
+--------+
| NOMPIL |
+--------+
| Benani |
+--------+
```

6. Avions à Marrakech ou capacité < 350 :

```
MariaDB [BD_AIR_MAROC]> SELECT NUMAV, NOMAV FROM AVION WHERE VILLE = 'Marrakech' OR CAPACITE < 350;
+-------+--------------+
| NUMAV | NOMAV        |
+-------+--------------+
|   100 | Boeing 737   |
|   101 | Airbus A320  |
|   103 | Airbus A350  |
|   104 | ATR 72       |
|   106 | Airbus A330  |
|   107 | Embraer 190  |
|   108 | Boeing 787   |
|   109 | Airbus A321  |
|   110 | Boeing 737   |
|   111 | Airbus A320  |
|   113 | Airbus A350  |
|   114 | ATR 72       |
|   115 | Boeing 777   |
|   116 | Airbus A330  |
|   117 | Embraer
190 |
|   118 | Boeing 787   |
|   119 | Airbus A321  |
|   120 | Boeing 737   |
|   121 | Airbus A320  |
|   123 | Airbus A350  |
|   124 | ATR 72       |
|   125 | Boeing 777   |
|   126 | Airbus A330  |
|   127 | Embraer 190  |
|   128 | Boeing 787   |
|   129 | Airbus A321  |
|   130 | Boeing 737   |
|   131 | Airbus A320  |
|   133 | Airbus A350  |
|   134 | ATR 72       |
|   135 | Boeing 777   |
|   136 | Airbus A330  |
|   137 | Embraer 190  |
|   138 | Boeing 787   |
|   139 | Airbus A321  |
|   140 | Boeing 737   |
|   141 | Airbus A320  |
|   143 | Airbus A350  |
|   144 | ATR 72       |
|   145 | Boeing 777   |
|   146 | Airbus A330  |
|   147 | Embraer
190 |
|   148 | Boeing 787   |
|   149 | Airbus A321  |
+-------+--------------+
44 rows in set (0.000 sec)
```

7. Pilotes pas en service :

```
MariaDB [BD_AIR_MAROC]> SELECT NUMPIL FROM PILOTES WHERE NUMPIL NOT IN (SELECT NUMPIL FROM VOL);
+--------+
| NUMPIL |
+--------+
|      1 |
|     46 |
|     47 |
|     48 |
|     49 |
+--------+
5 rows in set (0.004 sec)
```

8. Vols au départ de Marrakech par pilotes de Meknès :

```
MariaDB [BD_AIR_MAROC]> SELECT NUMVOL FROM VOL V JOIN PILOTES P ON V.NUMPIL = P.NUMPIL WHERE V.VILLE_DEP =
    -> 'Marrakech' AND P.VILLE = 'Meknès';
Empty set (0.000 sec)
```

9. Vols par avion non localisé à Marrakech :

```
MariaDB [BD_AIR_MAROC]> SELECT NUMVOL FROM VOL V JOIN AVION A ON V.NUMAV = A.NUMAV WHERE A.VILLE <> 'Marrakech';
+--------+
| NUMVOL |
+--------+
| V101   |
| V103   |
| V109   |
| V110   |
| V108   |
| V111   |
| V107   |
| V112   |
| V102   |
| V113   |
| V114   |
| V115   |
| V116   |
| V104   |
| V117   |
| V118   |
| V119   |
| V120   |
| V121   |
| V150   |
| V123   |
| V124   |
| V125   |
| V106   |
| V126   |
| V127   |
| V128   |
| V129   |
| V130   |
| V131   |
| V132   |
| V133   |
| V134   |
| V135   |
| V137   |
| V138   |
| V139   |
| V140   |
| V141   |
| V142   |
| V143   |
| V144   |
| V105   |
| V145   |
| V146   |
| V147   |
| V148   |
| V149   |
+--------+
48 rows in set (0.001 sec)
```

10. Villes après Guelmim :

```
MariaDB [BD_AIR_MAROC]> SELECT DISTINCT VILLE_ARR FROM VOL WHERE VILLE_DEP IN (SELECT VILLE_ARR FROM VOL WHERE
    -> VILLE_DEP = 'Guelmim');
+-----------+
| VILLE_ARR |
+-----------+
| Marrakech |
| Tanger    |
| Oujda     |
| Rabat     |
| Fes       |
| Meknes    |
| Safi      |
| Agadir    |
| Kenitra   |
| Tetouan   |
+-----------+
10 rows in set (0.001 sec)
```

11. Pilotes et numéros de vols :

```
MariaDB [BD_AIR_MAROC]> SELECT NOMPIL, NUMVOL FROM PILOTES P JOIN VOL V ON P.NUMPIL = V.NUMPIL;
+----------+--------+
| NOMPIL   | NUMVOL |
+----------+--------+
| Benani   | V108   |
| Benani   | V109   |
| Chraibi  | V110   |
| Drissi   | V111   |
| El Fassi | V104   |
| Ghazali  | V112   |
| Haddi    | V113   |
| Idrissi  | V114   |
| Jebli    | V115   |
| Kabbaj   | V102   |
| Kabbaj   | V116   |
| Kabbaj   | V150   |
| Lahlou   | V106   |
| Lahlou   | V117   |
| Mansouri | V101   |
| Mansouri | V103   |
| Naceri   | V118   |
| Ouazzani | V119   |
| Qasmi    | V120   |
| Radi     | V121   |
| Saidi    | V122   |
| Tazi     | V123   |
| Uali     | V124   |
| Vafi     | V125   |
| Wadi     | V126   |
| Xali     | V127   |
| Yacoubi  | V128   |
| Zouhair  | V129   |
| Amrani   | V107   |
| Bekkali  | V130   |
| Cheddadi | V131   |
| Dahbi    | V132   |
| Filali   | V133   |
| Gouni    | V134   |
| Habibi   | V135   |
| Iraqi    | V136   |
| Jouahri  | V137   |
| Kadir    | V138   |
| Lamrani  | V139   |
| Mernissi | V140   |
| Naji     | V141   |
| Omari    | V142   |
| Perrin   | V143   |
| Rami     | V144   |
| Slaoui   | V145   |
| Toumi    | V146   |
| Utmani   | V147   |
| Vadi     | V148   |
| Wafi     | V149   |
| Chafi    | V105   |
+----------+--------+
50 rows in set (0.000 sec)
```

12. Noms pilotes et noms avions pilotés :

```
MariaDB [BD_AIR_MAROC]> SELECT DISTINCT NOMPIL, NOMAV FROM PILOTES P JOIN VOL V ON P.NUMPIL = V.NUMPIL JOIN AVION
    -> A ON V.NUMAV = A.NUMAV;
+----------+--------------+
| NOMPIL   | NOMAV        |
+----------+--------------+
| Mansouri | Boeing 747   |
| Kabbaj   | Boeing 777   |
| El Fassi | Boeing 737   |
| Chafi    | Boeing 737   |
| Lahlou   | Boeing 737   |
| Amrani   | ATR 72       |
| Benani   | Airbus A350  |
| Benani   | Boeing 747   |
| Chraibi  | Boeing 747   |
| Drissi   | Airbus A350  |
| Ghazali  | ATR 72       |
| Haddi    | Airbus A330  |
| Idrissi  | Embraer 190  |
| Jebli    | Boeing 787   |
| Kabbaj   | Airbus A321  |
| Lahlou   | Airbus A320  |
| Naceri   | Boeing 747   |
| Ouazzani | Airbus A350  |
| Qasmi    | ATR 72       |
| Radi     | Boeing 777   |
| Saidi    | Airbus A330  |
| Tazi     | Embraer
190 |
| Uali     | Boeing 787   |
| Vafi     | Airbus A321  |
| Wadi     | Airbus A320  |
| Xali     | Boeing 747   |
| Yacoubi  | Airbus A350  |
| Zouhair  | ATR 72       |
| Bekkali  | Boeing 777   |
| Cheddadi | Airbus A330  |
| Dahbi    | Embraer 190  |
| Filali   | Boeing 787   |
| Gouni    | Airbus A321  |
| Habibi   | Boeing 737   |
| Iraqi    | Airbus A320  |
| Jouahri  | Boeing 747   |
| Kadir    | Airbus A350  |
| Lamrani  | ATR 72       |
| Mernissi | Boeing 777   |
| Naji     | Airbus A330  |
| Omari    | Embraer 190  |
| Perrin   | Boeing 787   |
| Rami     | Airbus A321  |
| Slaoui   | Boeing 737   |
| Toumi    | Airbus A320  |
| Utmani   | Boeing 747   |
| Vadi     | Airbus A350  |
| Wafi     | ATR 72       |
+----------+--------------+
48 rows in set (0.001 sec)
```

13. Vols avec nom du pilote :

```
MariaDB [BD_AIR_MAROC]> SELECT NUMVOL, VILLE_DEP, VILLE_ARR, NOMPIL FROM VOL V JOIN PILOTES P ON V.NUMPIL = P.NUMPIL;
+--------+------------+------------+----------+
| NUMVOL | VILLE_DEP  | VILLE_ARR  | NOMPIL   |
+--------+------------+------------+----------+
| V101   | Casablanca | Marrakech  | Mansouri |
| V102   | Marrakech  | Casablanca | Kabbaj   |
| V103   | Agadir     | Tanger     | Mansouri |
| V104   | Fes        | Oujda      | El Fassi |
| V105   | Guelmim    | Meknes     | Chafi    |
| V106   | Casablanca | Rabat      | Lahlou   |
| V107   | Tetouan    | Casablanca | Amrani   |
| V108   | Meknes     | Fes        | Benani   |
| V109   | Fes        | Meknes     | Benani   |
| V110   | Guelmim    | Agadir     | Chraibi  |
| V111   | Casablanca | Tanger     | Drissi   |
| V112   | Fes        | Marrakech  | Ghazali  |
| V113   | Tanger     | Oujda      | Haddi    |
| V114   | Agadir     | Safi       | Idrissi  |
| V115   | Oujda      | Nador      | Jebli    |
| V116   | Kenitra    | Settat     | Kabbaj   |
| V117   | Tetouan    | Rabat      | Lahlou   |
| V118   | El Jadida  | Casablanca | Naceri   |
| V119   | Nador      | Fes        | Ouazzani |
| V120   | Settat     | Meknes     | Qasmi    |
| V121   | Marrakech  | Guelmim    | Radi     |
| V122   | Meknes     | Tanger     | Saidi    |
| V123   | Guelmim    | Casablanca | Tazi     |
| V124   | Casablanca | Agadir     | Uali     |
| V125   | Rabat      | Oujda      | Vafi     |
| V126   | Fes        | Kenitra    | Wadi     |
| V127   | Tanger     | Tetouan    | Xali     |
| V128   | Agadir     | Safi       | Yacoubi  |
| V129   | Oujda      | El Jadida  | Zouhair  |
| V130   | Tetouan    | Nador      | Bekkali  |
| V131   | Safi       | Settat     | Cheddadi |
| V132   | El Jadida  | Marrakech  | Dahbi    |
| V133   | Nador      | Meknes     | Filali   |
| V134   | Settat     | Guelmim    | Gouni    |
| V135   | Marrakech  | Casablanca | Habibi   |
| V136   | Meknes     | Rabat      | Iraqi    |
| V137   | Guelmim    | Fes        | Jouahri  |
| V138   | Casablanca | Tanger     | Kadir    |
| V139   | Rabat      | Agadir     | Lamrani  |
| V140   | Fes        | Oujda      | Mernissi |
| V141   | Tanger     | Kenitra    | Naji     |
| V142   | Agadir     | Tetouan    | Omari    |
| V143   | Oujda      | Safi       | Perrin   |
| V144   | Kenitra    | El Jadida  | Rami     |
| V145   | Tetouan    | Nador      | Slaoui   |
| V146   | Safi       | Settat     | Toumi    |
| V147   | El Jadida  | Marrakech  | Utmani   |
| V148   | Nador      | Meknes     | Vadi     |
| V149   | Settat     | Guelmim    | Wafi     |
| V150   | Marrakech  | Casablanca | Kabbaj   |
+--------+------------+------------+----------+
50 rows in set (0.000 sec)
```

14. Avions au départ de Casablanca :

```
MariaDB [BD_AIR_MAROC]> SELECT DISTINCT NOMAV FROM AVION A JOIN VOL V ON A.NUMAV = V.NUMAV WHERE V.VILLE_DEP = 'Casablanca';
+-------------+
| NOMAV       |
+-------------+
| Boeing 747  |
| Boeing 737  |
| Airbus A350 |
| Boeing 787  |
+-------------+
4 rows in set (0.000 sec)
```

15. Pilotes d'avions de capacité > 300 :

```
MariaDB [BD_AIR_MAROC]> SELECT DISTINCT NOMPIL FROM PILOTES P JOIN VOL V ON P.NUMPIL = V.NUMPIL JOIN AVION A ON
    -> V.NUMAV = A.NUMAV WHERE A.CAPACITE > 300;
+----------+
| NOMPIL   |
+----------+
| Mansouri |
| Kabbaj   |
| Benani   |
| Chraibi  |
| Drissi   |
| Jebli    |
| Naceri   |
| Ouazzani |
| Radi     |
| Xali     |
| Bekkali  |
| Jouahri  |
| Mernissi |
| Utmani   |
+----------+
14 rows in set (0.002 sec)
```

16. Pilotes et villes d'arrivée :

```
MariaDB [BD_AIR_MAROC]> SELECT NOMPIL, VILLE_ARR FROM PILOTES P JOIN VOL V ON P.NUMPIL = V.NUMPIL;
+----------+------------+
| NOMPIL   | VILLE_ARR  |
+----------+------------+
| Mansouri | Marrakech  |
| Kabbaj   | Casablanca |
| Mansouri | Tanger     |
| El Fassi | Oujda      |
| Chafi    | Meknes     |
| Lahlou   | Rabat      |
| Amrani   | Casablanca |
| Benani   | Fes        |
| Benani   | Meknes     |
| Chraibi  | Agadir     |
| Drissi   | Tanger     |
| Ghazali  | Marrakech  |
| Haddi    | Oujda      |
| Idrissi  | Safi       |
| Jebli    | Nador      |
| Kabbaj   | Settat     |
| Lahlou   | Rabat      |
| Naceri   | Casablanca |
| Ouazzani | Fes        |
| Qasmi    | Meknes     |
| Radi     | Guelmim    |
| Saidi    | Tanger     |
| Tazi     | Casablanca |
| Uali     | Agadir     |
| Vafi     | Oujda      |
| Wadi     | Kenitra    |
| Xali     | Tetouan    |
| Yacoubi  | Safi       |
| Zouhair  | El Jadida  |
| Bekkali  | Nador      |
| Cheddadi | Settat     |
| Dahbi    | Marrakech  |
| Filali   | Meknes     |
| Gouni    | Guelmim    |
| Habibi   | Casablanca |
| Iraqi    | Rabat      |
| Jouahri  | Fes        |
| Kadir    | Tanger     |
| Lamrani  | Agadir     |
| Mernissi | Oujda      |
| Naji     | Kenitra    |
| Omari    | Tetouan    |
| Perrin   | Safi       |
| Rami     | El Jadida  |
| Slaoui   | Nador      |
| Toumi    | Settat     |
| Utmani   | Marrakech  |
| Vadi     | Meknes     |
| Wafi     | Guelmim    |
| Kabbaj   | Casablanca |
+----------+------------+
50 rows in set (0.000 sec)
```

17. Jointure complète des 3 tables :

```
MariaDB [BD_AIR_MAROC]> SELECT * FROM VOL V JOIN PILOTES P ON V.NUMPIL = P.NUMPIL JOIN AVION A ON V.NUMAV =
    -> A.NUMAV;
+--------+--------+--------+------------+------------+---------+---------+--------+---------+------------+---------+--------+-----------------+----------+------------+
| Numvol | NUMPIL | NUMAV  | Ville_dep  | Ville_arr  | H_dep   | H_arr   | NUMPIL | NOMPIL  | Ville      | Salaire | NUMAV  | Nomav           | Capacite | Ville      |
+--------+--------+--------+------------+------------+---------+---------+--------+---------+------------+---------+--------+-----------------+----------+------------+
| V101   |     12 |    102 | Casablanca | Marrakech  | 08:00:00 | 09:00:00 |     12 | Mansouri | Safi       | 13000.00 |    102 | Boeing 747      |      400 | Rabat      |
| V102   |     10 |    105 | Marrakech  | Casablanca | 10:30:00 | 11:30:00 |     10 | Kabbaj  | Kenitra    | 30000.00 |    105 | Boeing 777      |      350 | Tanger     |
| V103   |     12 |    102 | Agadir     | Tanger     | 14:00:00 | 15:45:00 |     12 | Mansouri | Safi       | 13000.00 |    102 | Boeing 747      |      400 | Rabat      |
| V104   |      5 |    110 | Fes        | Oujda      | 07:15:00 | 08:00:00 |      5 | El Fassi | Rabat      | 29500.00 |    110 | Boeing 737      |      185 | Tetouan    |
| V105   |     50 |    140 | Guelmim    | Meknes     | 16:00:00 | 18:30:00 |     50 | Chafi   | Rabat      | 15000.00 |    140 | Boeing 737      |      200 | Tetouan    |
| V106   |     11 |    120 | Casablanca | Rabat      | 19:00:00 | 19:45:00 |     11 | Lahlou  | Tetouan    | 12000.00 |    120 | Boeing 737      |      190 | Tanger     |
| V107   |     25 |    104 | Tetouan    | Casablanca | 11:00:00 | 12:15:00 |     25 | Amrani  | Kenitra    | 27500.00 |    104 | ATR 72          |      160 | Agadir     |
| V108   |      2 |    103 | Meknes     | Fes        | 08:30:00 | 09:00:00 |      2 | Benani  | Meknes     | 30000.00 |    103 | Airbus A350     |      320 | Fes        |
| V109   |      2 |    102 | Fes        | Meknes     | 10:00:00 | 10:30:00 |      2 | Benani  | Meknes     | 30000.00 |    102 | Boeing 747      |      400 | Rabat      |
| V110   |      3 |    102 | Guelmim    | Agadir     | 13:00:00 | 14:00:00 |      3 | Chraibi | Guelmim    | 28000.00 |    102 | Boeing 747      |      400 | Rabat      |
| V111   |      4 |    103 | Casablanca | Tanger     | 15:00:00 | 16:30:00 |      4 | Drissi  | Casablanca | 13500.00 |    103 | Airbus A350     |      320 | Fes        |
| V112   |      6 |    104 | Fes        | Marrakech  | 09:00:00 | 10:30:00 |      6 | Ghazali | Fes        | 12000.00 |    104 | ATR 72          |      160 | Agadir     |
| V113   |      7 |    106 | Tanger     | Oujda      | 12:00:00 | 13:15:00 |      7 | Haddi   | Tanger     | 25500.00 |    106 | Airbus A330     |      280 | Meknes     |
| V114   |      8 |    107 | Agadir     | Safi       | 17:00:00 | 18:00:00 |      8 | Idrissi | Agadir     | 12000.00 |    107 | Embraer 190     |      170 | Oujda      |
| V115   |      9 |    108 | Oujda      | Nador      | 08:00:00 | 08:45:00 |      9 | Jebli   | Oujda      | 12000.00 |    108 | Boeing 787      |      310 | Guelmim    |
| V116   |     10 |    109 | Kenitra    | Settat     | 14:00:00 | 15:00:00 |     10 | Kabbaj  | Kenitra    | 30000.00 |    109 | Airbus A321     |      200 | Kenitra    |
| V117   |     11 |    111 | Tetouan    | Rabat      | 10:00:00 | 11:15:00 |     11 | Lahlou  | Tetouan    | 12000.00 |    111 | Airbus A320     |      155 | Safi       |
| V118   |     13 |    112 | El Jadida  | Casablanca | 16:00:00 | 17:00:00 |     13 | Naceri  | El Jadida  | 19500.00 |    112 | Boeing 747      |      390 | El Jadida  |
| V119   |     14 |    113 | Nador      | Fes        | 07:00:00 | 08:00:00 |     14 | Ouazzani | Nador      | 13200.00 |    113 | Airbus A350     |      310 | Nador      |
| V120   |     15 |    114 | Settat     | Meknes     | 18:00:00 | 19:30:00 |     15 | Qasmi   | Settat     | 26400.00 |    114 | ATR 72          |      165 | Settat     |
| V121   |     16 |    115 | Marrakech  | Guelmim    | 09:30:00 | 10:45:00 |     16 | Radi    | Marrakech  | 28000.00 |    115 | Boeing 777      |      340 | Casablanca |
| V122   |     17 |    116 | Meknes     | Tanger     | 11:00:00 | 12:30:00 |     17 | Saidi   | Meknes     | 13900.00 |    116 | Airbus A330     |      270 | Marrakech  |
| V123   |     18 |    117 | Guelmim    | Casablanca | 15:00:00 | 16:45:00 |     18 | Tazi    | Guelmim    | 18200.00 |    117 | Embraer
190 |      175 | Rabat      |
| V124   |     19 |    118 | Casablanca | Agadir     | 08:00:00 | 09:30:00 |     19 | Uali    | Casablanca | 10800.00 |    118 | Boeing 787      |      300 | Fes        |
| V125   |     20 |    119 | Rabat      | Oujda      | 13:00:00 | 14:30:00 |     20 | Vafi    | Rabat      | 24500.00 |    119 | Airbus A321     |      210 | Agadir     |
| V126   |     21 |    121 | Fes        | Kenitra    | 10:00:00 | 11:00:00 |     21 | Wadi    | Fes        | 19200.00 |    121 | Airbus A320     |      160 | Meknes     |
| V127   |     22 |    122 | Tanger     | Tetouan    | 14:00:00 | 14:45:00 |     22 | Xali    | Tanger     | 30000.00 |    122 | Boeing 747      |      380 | Oujda      |
| V128   |     23 |    123 | Agadir     | Safi       | 17:00:00 | 18:15:00 |     23 | Yacoubi | Agadir     | 18200.00 |    123 | Airbus A350     |      300 | Guelmim    |
| V129   |     24 |    124 | Oujda      | El Jadida  | 09:00:00 | 10:30:00 |     24 | Zouhair | Oujda      | 30000.00 |    124 | ATR 72          |      170 | Kenitra    |
| V130   |     26 |    125 | Tetouan    | Nador      | 12:00:00 | 13:00:00 |     26 | Bekkali | Tetouan    | 21200.00 |    125 | Boeing 777      |      330 | Tetouan    |
| V131   |     27 |    126 | Safi       | Settat     | 15:00:00 | 16:15:00 |     27 | Cheddadi | Safi       | 30000.00 |    126 | Airbus A330     |      260 | Safi       |
| V132   |     28 |    127 | El Jadida  | Marrakech  | 08:00:00 | 09:45:00 |     28 | Dahbi   | El Jadida  | 29700.00 |    127 | Embraer 190     |      180 | El Jadida  |
| V133   |     29 |    128 | Nador      | Meknes     | 11:00:00 | 12:45:00 |     29 | Filali  | Nador      | 23400.00 |    128 | Boeing 787      |      290 | Nador      |
| V134   |     30 |    129 | Settat     | Guelmim    | 16:00:00 | 17:45:00 |     30 | Gouni   | Settat     | 16700.00 |    129 | Airbus A321     |      220 | Settat     |
| V135   |     31 |    130 | Marrakech  | Casablanca | 07:00:00 | 08:00:00 |     31 | Habibi  | Marrakech  | 30000.00 |    130 | Boeing 737      |      195 | Casablanca |
| V136   |     32 |    131 | Meknes     | Rabat      | 10:00:00 | 11:15:00 |     32 | Iraqi   | Meknes     | 12000.00 |    131 | Airbus A320     |      165 | Marrakech  |
| V137   |     33 |    132 | Guelmim    | Fes        | 14:00:00 | 15:45:00 |     33 | Jouahri | Guelmim    | 18400.00 |    132 | Boeing 747      |      370 | Rabat      |
| V138   |     34 |    133 | Casablanca | Tanger     | 18:00:00 | 19:30:00 |     34 | Kadir   | Casablanca | 20500.00 |    133 | Airbus A350     |      290 | Fes        |
| V139   |     35 |    134 | Rabat      | Agadir     | 09:00:00 | 10:45:00 |     35 | Lamrani | Rabat      | 24800.00 |    134 | ATR 72          |      175 | Agadir     |
| V140   |     36 |    135 | Fes        | Oujda      | 12:00:00 | 13:00:00 |     36 | Mernissi | Fes        | 12000.00 |    135 | Boeing 777      |      320 | Tanger     |
| V141   |     37 |    136 | Tanger     | Kenitra    | 15:00:00 | 16:00:00 |     37 | Naji    | Tanger     | 30000.00 |    136 | Airbus A330     |      250 | Meknes     |
| V142   |     38 |    137 | Agadir     | Tetouan    | 08:00:00 | 10:00:00 |     38 | Omari   | Agadir     | 12000.00 |    137 | Embraer 190     |      185 | Oujda      |
| V143   |     39 |    138 | Oujda      | Safi       | 11:00:00 | 13:00:00 |     39 | Perrin  | Oujda      | 26000.00 |    138 | Boeing 787      |      280 | Guelmim    |
| V144   |     40 |    139 | Kenitra    | El Jadida  | 16:00:00 | 17:30:00 |     40 | Rami    | Kenitra    | 17800.00 |    139 | Airbus A321     |      230 | Kenitra    |
| V145   |     41 |    140 | Tetouan    | Nador      | 07:30:00 | 08:30:00 |     41 | Slaoui  | Tetouan    | 29400.00 |    140 | Boeing 737      |      200 | Tetouan    |
| V146   |     42 |    141 | Safi       | Settat     | 14:00:00 | 15:15:00 |     42 | Toumi   | Safi       | 30000.00 |    141 | Airbus A320     |      170 | Safi       |
| V147   |     43 |    142 | El Jadida  | Marrakech  | 19:00:00 | 20:45:00 |     43 | Utmani  | El Jadida  | 19900.00 |    142 | Boeing 747      |      360 | El Jadida  |
| V148   |     44 |    143 | Nador      | Meknes     | 10:00:00 | 11:45:00 |     44 | Vadi    | Nador      | 23600.00 |    143 | Airbus A350     |      280 | Nador      |
| V149   |     45 |    144 | Settat     | Guelmim    | 13:00:00 | 14:45:00 |     45 | Wafi    | Settat     | 27000.00 |    144 | ATR 72          |      180 | Settat     |
| V150   |     10 |    115 | Marrakech  | Casablanca | 21:00:00 | 22:00:00 |     10 | Kabbaj  | Kenitra    | 30000.00 |    115 | Boeing 777      |      340 | Casablanca |
+--------+--------+--------+------------+------------+---------+---------+--------+---------+------------+---------+--------+-----------------+----------+------------+
50 rows in set (0.002 sec)
```

18. Pilotes avec salaire > moyenne :

```
MariaDB [BD_AIR_MAROC]> SELECT NOMPIL FROM PILOTES WHERE SALAIRE > (SELECT AVG(SALAIRE) FROM PILOTES);
+----------+
| NOMPIL   |
+----------+
| Benani   |
| Chraibi  |
| El Fassi |
| Haddi    |
| Kabbaj   |
| Qasmi    |
| Radi     |
| Vafi     |
| Xali     |
| Zouhair  |
| Amrani   |
| Cheddadi |
| Dahbi    |
| Filali   |
| Habibi   |
| Lamrani  |
| Naji     |
| Perrin   |
| Slaoui   |
| Toumi    |
| Vadi     |
| Wafi     |
| Younes   |
| Bari     |
+----------+
24 rows in set (0.001 sec)
```

19. Avions capacité > avion 101 :

```
MariaDB [BD_AIR_MAROC]> SELECT * FROM AVION WHERE CAPACITE > (SELECT CAPACITE FROM AVION WHERE NUMAV = 101);
+--------+--------------+----------+------------+
| NUMAV | Nomav        | Capacite | Ville      |
+--------+--------------+----------+------------+
|    100 | Boeing 737   |      180 | Casablanca |
|    102 | Boeing 747   |      400 | Rabat      |
|    103 | Airbus A350  |      320 | Fes        |
|    104 | ATR 72       |      160 | Agadir     |
|    105 | Boeing 777   |      350 | Tanger     |
|    106 | Airbus A330  |      280 | Meknes     |
|    107 | Embraer 190  |      170 | Oujda      |
|    108 | Boeing 787   |      310 | Guelmim    |
|    109 | Airbus A321  |      200 | Kenitra    |
|    110 | Boeing 737   |      185 | Tetouan    |
|    111 | Airbus A320  |      155 | Safi       |
|    112 | Boeing 747   |      390 | El Jadida  |
|    113 | Airbus A350  |      310 | Nador      |
|    114 | ATR 72       |      165 | Settat     |
|    115 | Boeing 777   |      340 | Casablanca |
|    116 | Airbus A330  |      270 | Marrakech  |
|    117 | Embraer
190 |      175 | Rabat      |
|    118 | Boeing 787   |      300 | Fes        |
|    119 | Airbus A321  |      210 | Agadir     |
|    120 | Boeing 737   |      190 | Tanger     |
|    121 | Airbus A320  |      160 | Meknes     |
|    122 | Boeing 747   |      380 | Oujda      |
|    123 | Airbus A350  |      300 | Guelmim    |
|    124 | ATR 72       |      170 | Kenitra    |
|    125 | Boeing 777   |      330 | Tetouan    |
|    126 | Airbus A330  |      260 | Safi       |
|    127 | Embraer 190  |      180 | El Jadida  |
|    128 | Boeing 787   |      290 | Nador      |
|    129 | Airbus A321  |      220 | Settat     |
|    130 | Boeing 737   |      195 | Casablanca |
|    131 | Airbus A320  |      165 | Marrakech  |
|    132 | Boeing 747   |      370 | Rabat      |
|    133 | Airbus A350  |      290 | Fes        |
|    134 | ATR 72       |      175 | Agadir     |
|    135 | Boeing 777   |      320 | Tanger     |
|    136 | Airbus A330  |      250 | Meknes     |
|    137 | Embraer 190  |      185 | Oujda      |
|    138 | Boeing 787   |      280 | Guelmim    |
|    139 | Airbus A321  |      230 | Kenitra    |
|    140 | Boeing 737   |      200 | Tetouan    |
|    141 | Airbus A320  |      170 | Safi       |
|    142 | Boeing 747   |      360 | El Jadida  |
|    143 | Airbus A350  |      280 | Nador      |
|    144 | ATR 72       |      180 | Settat     |
|    145 | Boeing 777   |      310 | Casablanca |
|    146 | Airbus A330  |      240 | Marrakech  |
|    147 | Embraer
190 |      190 | Rabat      |
|    148 | Boeing 787   |      270 | Fes        |
|    149 | Airbus A321  |      240 | Agadir     |
+--------+--------------+----------+------------+
49 rows in set (0.001 sec)
```

20. Pilotes sans aucun vol :

```
MariaDB [BD_AIR_MAROC]> SELECT NOMPIL FROM PILOTES WHERE NUMPIL NOT IN (SELECT NUMPIL FROM VOL);
+--------+
| NOMPIL |
+--------+
| Alami  |
| Younes |
| Ziani  |
| Abidi  |
| Bari   |
+--------+
5 rows in set (0.001 sec)
```

21. Villes de départ des pilotes de Marrakech :

```
MariaDB [BD_AIR_MAROC]> SELECT DISTINCT VILLE_DEP FROM VOL V JOIN PILOTES P ON V.NUMPIL = P.NUMPIL WHERE P.VILLE =
    -> 'Marrakech';
+-----------+
| VILLE_DEP |
+-----------+
| Marrakech |
+-----------+
1 row in set (0.001 sec)
```

22. Vue vols avec noms pilote/avion :

```
MariaDB [BD_AIR_MAROC]> CREATE VIEW VUE_VOL_COMPLET AS SELECT V.*, P.NOMPIL, A.NOMAV FROM VOL V JOIN PILOTES P ON
    -> V.NUMPIL = P.NUMPIL JOIN AVION A ON V.NUMAV = A.NUMAV;
Query OK, 0 rows affected (0.007 sec)
```

23. Vue pilotes salaire > 15000 :

```
MariaDB [BD_AIR_MAROC]> CREATE VIEW VUE_PILOTES_RICHES AS SELECT * FROM PILOTES WHERE SALAIRE > 15000;
Query OK, 0 rows affected (0.010 sec)
```

24. Afficher la vue :

```
MariaDB [BD_AIR_MAROC]> SELECT * FROM VUE_PILOTES_RICHES;
+--------+----------+------------+----------+
| NUMPIL | NOMPIL   | Ville      | Salaire  |
+--------+----------+------------+----------+
|      2 | Benani   | Meknes     | 30000.00 |
|      3 | Chraibi  | Guelmim    | 28000.00 |
|      5 | El Fassi | Rabat      | 29500.00 |
|      7 | Haddi    | Tanger     | 25500.00 |
|     10 | Kabbaj   | Kenitra    | 30000.00 |
|     13 | Naceri   | El Jadida  | 19500.00 |
|     15 | Qasmi    | Settat     | 26400.00 |
|     16 | Radi     | Marrakech  | 28000.00 |
|     18 | Tazi     | Guelmim    | 18200.00 |
|     20 | Vafi     | Rabat      | 24500.00 |
|     21 | Wadi     | Fes        | 19200.00 |
|     22 | Xali     | Tanger     | 30000.00 |
|     23 | Yacoubi  | Agadir     | 18200.00 |
|     24 | Zouhair  | Oujda      | 30000.00 |
|     25 | Amrani   | Kenitra    | 27500.00 |
|     26 | Bekkali  | Tetouan    | 21200.00 |
|     27 | Cheddadi | Safi       | 30000.00 |
|     28 | Dahbi    | El Jadida  | 29700.00 |
|     29 | Filali   | Nador      | 23400.00 |
|     30 | Gouni    | Settat     | 16700.00 |
|     31 | Habibi   | Marrakech  | 30000.00 |
|     33 | Jouahri  | Guelmim    | 18400.00 |
|     34 | Kadir    | Casablanca | 20500.00 |
|     35 | Lamrani  | Rabat      | 24800.00 |
|     37 | Naji     | Tanger     | 30000.00 |
|     39 | Perrin   | Oujda      | 26000.00 |
|     40 | Rami     | Kenitra    | 17800.00 |
|     41 | Slaoui   | Tetouan    | 29400.00 |
|     42 | Toumi    | Safi       | 30000.00 |
|     43 | Utmani   | El Jadida  | 19900.00 |
|     44 | Vadi     | Nador      | 23600.00 |
|     45 | Wafi     | Settat     | 27000.00 |
|     46 | Younes   | Marrakech  | 30000.00 |
|     48 | Abidi    | Guelmim    | 18600.00 |
|     49 | Bari     | Casablanca | 30000.00 |
+--------+----------+------------+----------+
35 rows in set (0.001 sec)
```

25. Supprimer une vue :

```
MariaDB [BD_AIR_MAROC]> DROP VIEW VUE_PILOTES_RICHES
    -> ;
Query OK, 0 rows affected (0.001 sec)
```

26. Vue vols départ Marrakech :

```
MariaDB [BD_AIR_MAROC]> CREATE VIEW VUE_VOL_KECH AS SELECT * FROM VOL WHERE VILLE_DEP = 'Marrakech';
Query OK, 0 rows affected (0.005 sec)
```

27. Créer user_air :

```
MariaDB [BD_AIR_MAROC]> CREATE USER 'user_air'@'localhost' IDENTIFIED BY 'password';
Query OK, 0 rows affected (0.005 sec)
```

28. Droit SELECT sur PILOTES :

```
MariaDB [BD_AIR_MAROC]> GRANT SELECT ON BD_AIR_MAROC.PILOTES TO 'user_air'@'localhost';
Query OK, 0 rows affected (0.009 sec)
```

29. Droit INSERT sur VOL :

```
MariaDB [BD_AIR_MAROC]> GRANT INSERT ON BD_AIR_MAROC.VOL TO 'user_air'@'localhost';
Query OK, 0 rows affected (0.003 sec)
```

30. Retirer DELETE sur AVION :

```
MariaDB [BD_AIR_MAROC]> GRANT INSERT ON BD_AIR_MAROC.AVION TO 'user_air'@'localhost';
Query OK, 0 rows affected (0.003 sec)

MariaDB [BD_AIR_MAROC]> REVOKE DELETE ON BD_AIR_MAROC.AVION FROM 'user_air'@'localhost';
Query OK, 0 rows affected (0.003 sec)
```

31. Tous les privilèges :

```
MariaDB [BD_AIR_MAROC]> GRANT ALL PRIVILEGES ON BD_AIR_MAROC.* TO 'user_air'@'localhost';
Query OK, 0 rows affected (0.004 sec)
```

32. Créer rôle gestion_vol :

```
MariaDB [BD_AIR_MAROC]> CREATE ROLE gestion_vol; GRANT SELECT, INSERT, UPDATE ON VOL TO gestion_vol;
Query OK, 0 rows affected (0.004 sec)

Query OK, 0 rows affected (0.002 sec)
```

33. Affecter le rôle :

```
MariaDB [BD_AIR_MAROC]> GRANT gestion_vol TO 'user_air'@'localhost'
    -> ;
Query OK, 0 rows affected (0.003 sec)
```